BiasAperture — Demographic Fairness Audit Dossier

Diagnostic Demographic Bias Evaluation & Regulatory Traceability Report

AUDITED MODEL
FairFace ResNet-34 Multi-Task Classifier

EVALUATION DATASET
FairFace Benchmark Dataset (N=10954)

PROTECTED AXIS
Gender

AUDIT DATE
2026-09-02 12:29:04 UTC

Diagnostic Scope & Non-Intervention Guarantee:

BiasAperture is engineered exclusively as an external diagnostic auditing, statistical significance testing, and regulatory traceability platform. It does **not** perform model retraining, in-processing debiasing, or synthetic image generation. In adherence to statistical integrity safeguards (NFR-003), demographic subgroups with fewer than 30 samples (*n < 30*) have point estimates suppressed to prevent unreliable compliance claims.

2 / 2	Female	DIR = 0.070	0
SUBGROUPS WITH SIGNIFICANT DISPARITY (P < 0.05)	LARGEST DEVIATION SUBGROUP	WORST HEADLINE METRIC	SUBGROUPS FLAGGED N < 30 (NFR-003)

1. Headline Fairness Metrics (Core Four — Cross-Group Overview)

Summary metrics evaluated across all eligible demographic strata (n ≥ 30 per NFR-003). Accompanied by 95% stratified BCa bootstrap confidence intervals and Pearson's χ^2 test of independence. *Note: each CI band is scaled to its own row — dot colour (red = CI excludes null), not bar width, indicates significance across rows.*

Metric Name	Point Estimate	95% Bootstrap CI	CI Band vs. Null	Significance (p-value)	Regulatory Clause
Demographic Parity Difference	0.923	[0.915, 0.930]		p=0.0000 p < 0.05	EU AI Act Art. 10(2)(f) / NIST Measure 2.11
Equalized Odds Difference	N/A (n < 30)	—	—	—	EU AI Act Art. 10(2)(g) / NIST Measure 2.11
Equal Opportunity Difference	N/A (n < 30)	—	—	—	EU AI Act Art. 10(3) / NIST Measure 2.11
Disparate Impact Ratio	0.039	[0.033, 0.044]		p=0.0000 p < 0.05	EEOC 4/5ths Rule / EU AI Act Annex IV §2(g)

2. Per-Subgroup Disparity Matrix

Detailed stratification across individual demographic groups, ordered by largest deviation from the null (no-disparity) point, most significant first. Subgroups below the NFR-003 sample threshold (n < 30) are strictly flagged, uncomputed to prevent small-sample estimation artifacts, and summarised separately below.

Subgroup	Sample Size (n)	DPD	EOD (Max-Gap)	EOP	DIR (Symmetric)	Deviation from Null (red = p < 0.05)	Data Integrity Status
Female	5162	0.488	—	—	0.070		Audited (Valid)
Male	5792	0.435	—	0.000	0.547		Audited (Valid)

3. Governance Documentation & Intended Use

> **EU AI Act (Regulation EU 2024/1689) Article 10 / Annex IV Alignment:** This audit dossier provides verifiable diagnostic documentation of demographic performance disparities. It does not perform model retraining or in-situ weight modifications.

Model Description: ImageNet-pretrained ResNet-34 terminating in an 18-unit linear head sliced for race (7), gender (2), and age (9) classification.

Evaluation Protocol: Standalone offline evaluation using dual-backend cross-validation (Fairlearn + AIF360) and 1,000-resample stratified BCa bootstrap estimation.
